Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)
Electronics and Telecommunication Engineering Department

Academic year 2025-2026

EXP-4B

Name: Chinmay Tawade

UID: 2024200132

Batch: B4

Aim: To design and configure a small computer network consisting of PCs, switches, and routers using Cisco Packet Tracer; to assign static IP addressing to all end devices and router interfaces; to configure internetwork routing between two separate LANs; and to verify end-to-end connectivity using the ping utility and the Simulation mode, for both a point-to-point (star) topology and a bus topology.

Software:

Cisco Packet Tracer 9.01

Theory:

Cisco Packet Tracer is a network simulation program that allows users to build, configure, and troubleshoot networks virtually without needing physical networking equipment. It can be used to work with devices such as computers, switches, and routers and observe how they communicate with each other.

A network can remain in a "down" state even when all the devices are physically connected and the link indicators are active. This usually happens when important network settings, such as IP addresses, subnet masks, default gateways, or routing information, have not been configured correctly. Static IP configuration requires these details to be entered manually on each end device. Similarly, router interfaces need suitable IP addresses and must be activated using the `no shutdown` command.

When two different LANs are linked through a serial WAN connection, additional routing configuration is required. Static routes are added to the routers so they know where to send packets when the destination network is not directly connected to them.

Sardar Patel Institute of Technology

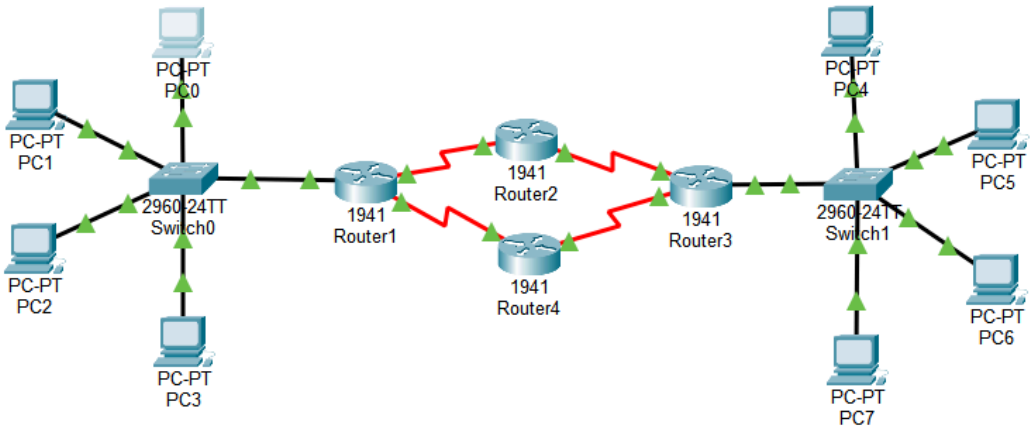
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)
Electronics and Telecommunication Engineering Department

Academic year 2025-2026

After the required addressing, interface, and routing configurations are completed, the network should be able to communicate properly. Connectivity can be checked using the `ping` command, while Packet Tracer's Simulation mode can be used to observe how packets move through the network from one device to another.

Circuit:



Name: Chinmay Tawade
UID: 2024200132

Batch: B4
Branch : EXTC

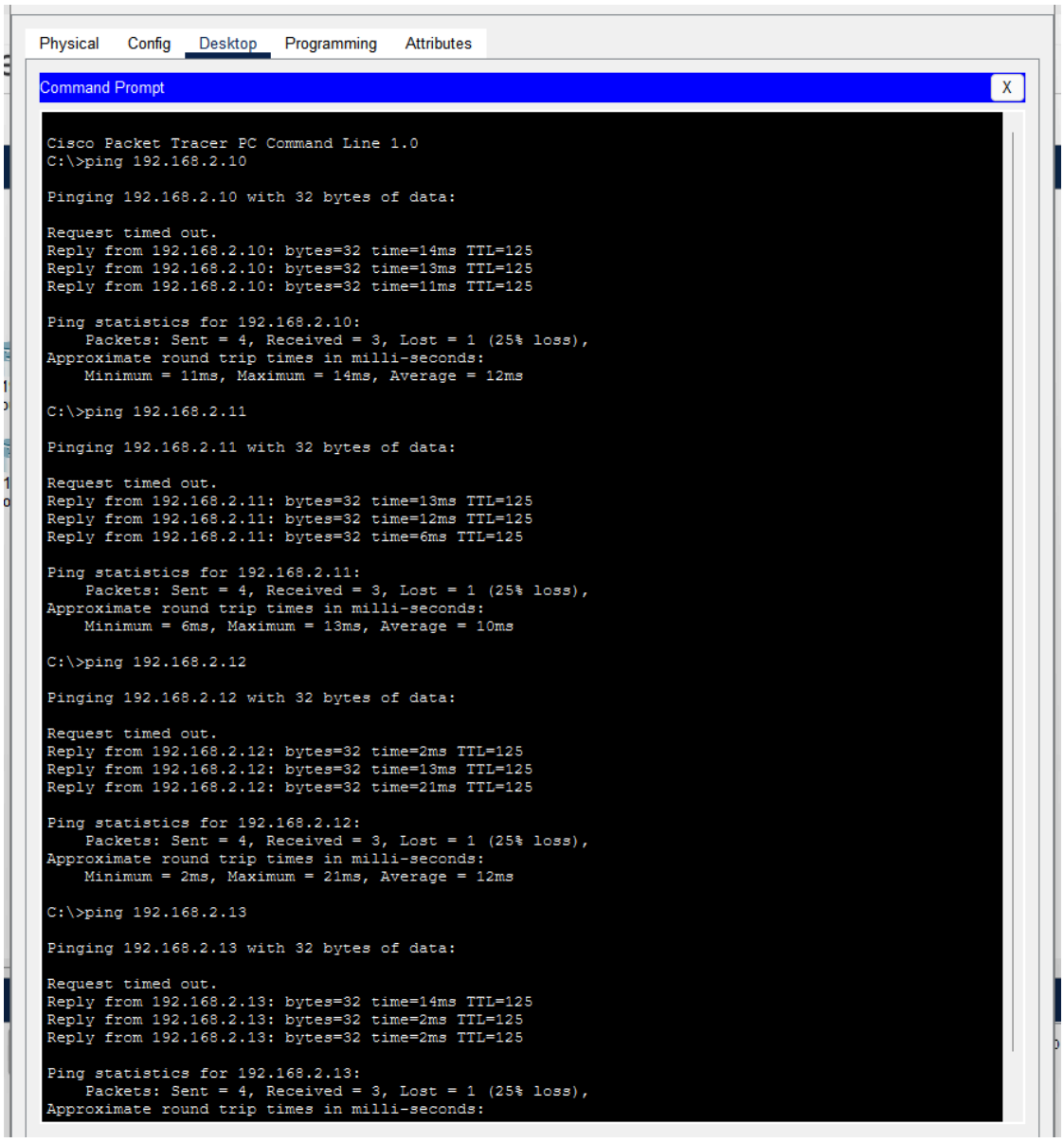
Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)
Electronics and Telecommunication Engineering Department

Academic year 2025-2026

Pinging Right side PCs from PCs on the left



Pinging left side PCs from PC on the right

Name: Chinmay Tawade
UID: 2024200132

Batch: B4
Branch : EXTC

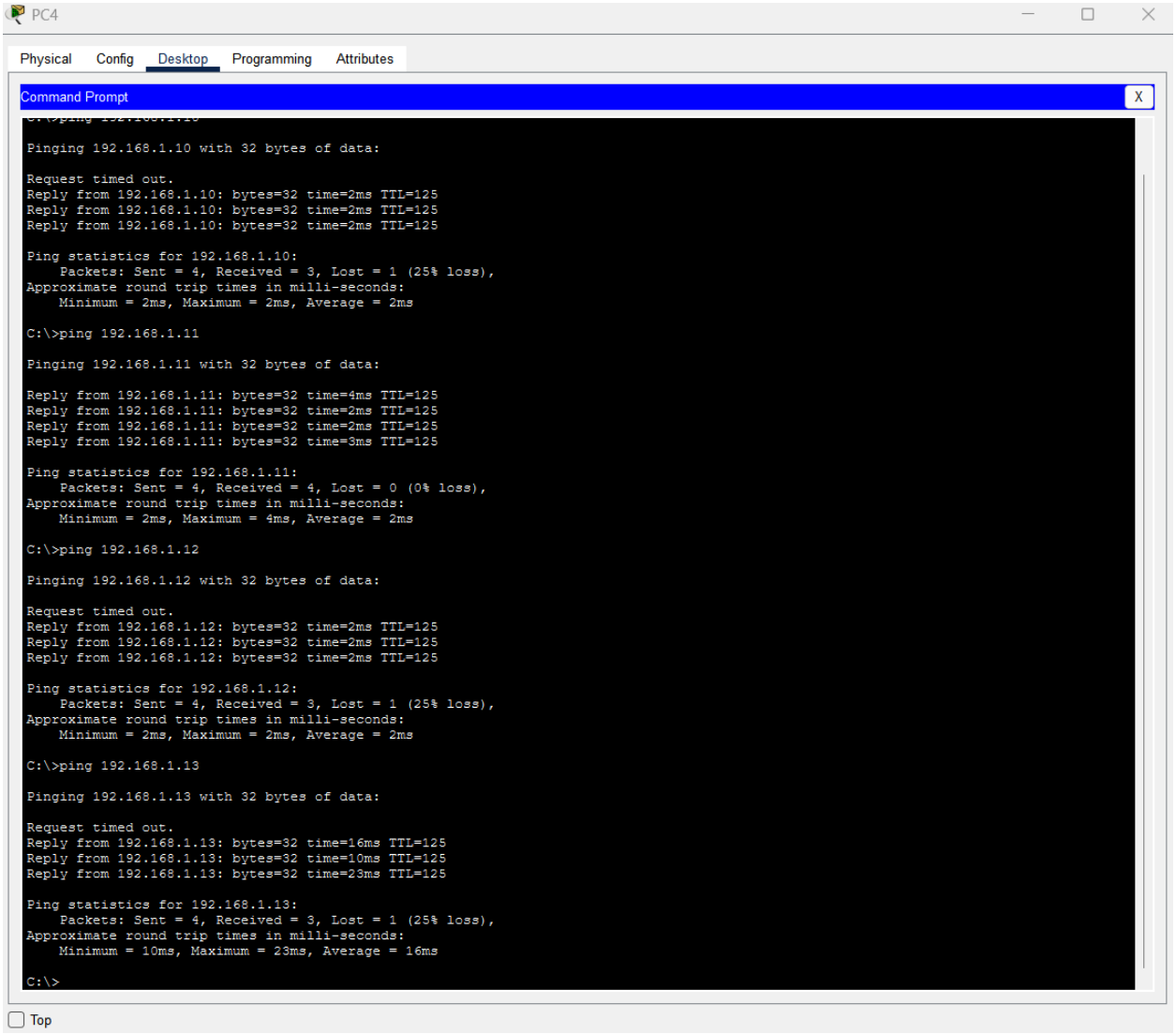
Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)

Electronics and Telecommunication Engineering Department

Academic year 2025-2026



Router Configurations

Name: Chinmay Tawade
UID: 2024200132

Batch: B4
Branch : EXTC

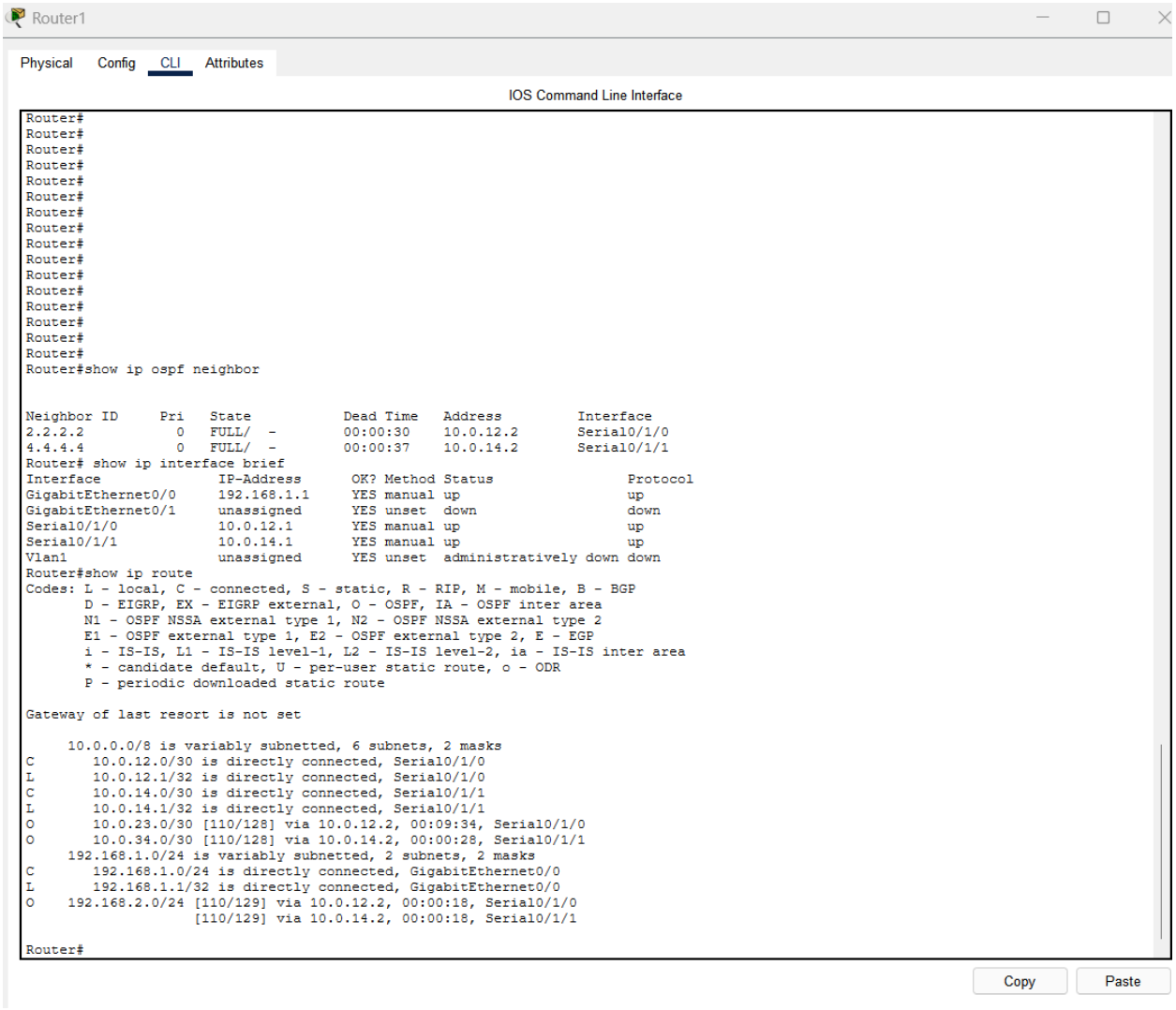
Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)

Electronics and Telecommunication Engineering Department

Academic year 2025-2026



Name: Chinmay Tawade

UID: 2024200132

Batch: B4

Branch : EXTC

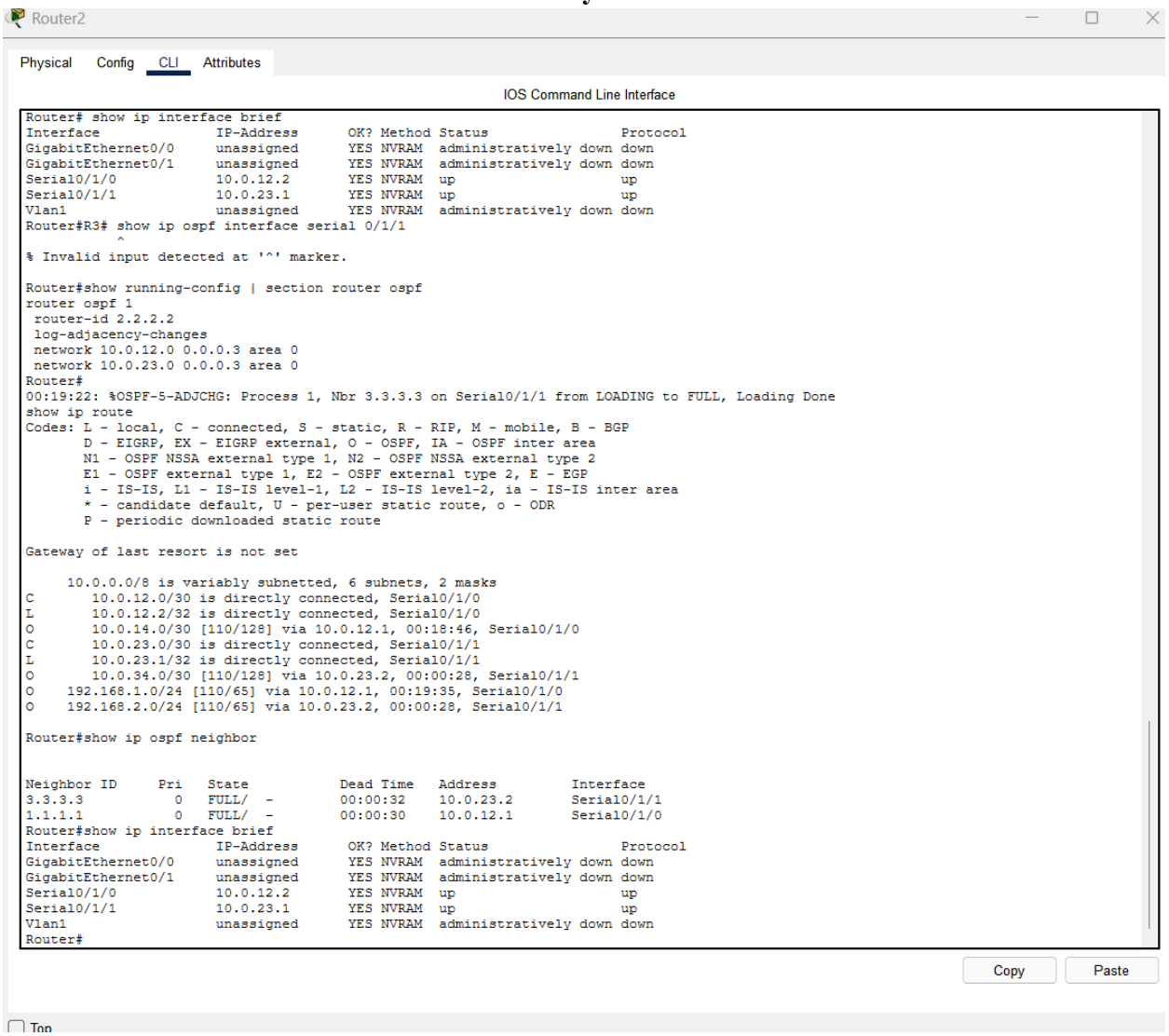
Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)

Electronics and Telecommunication Engineering Department

Academic year 2025-2026



Name: Chinmay Tawade

UID: 2024200132

Batch: B4

Branch : EXTC

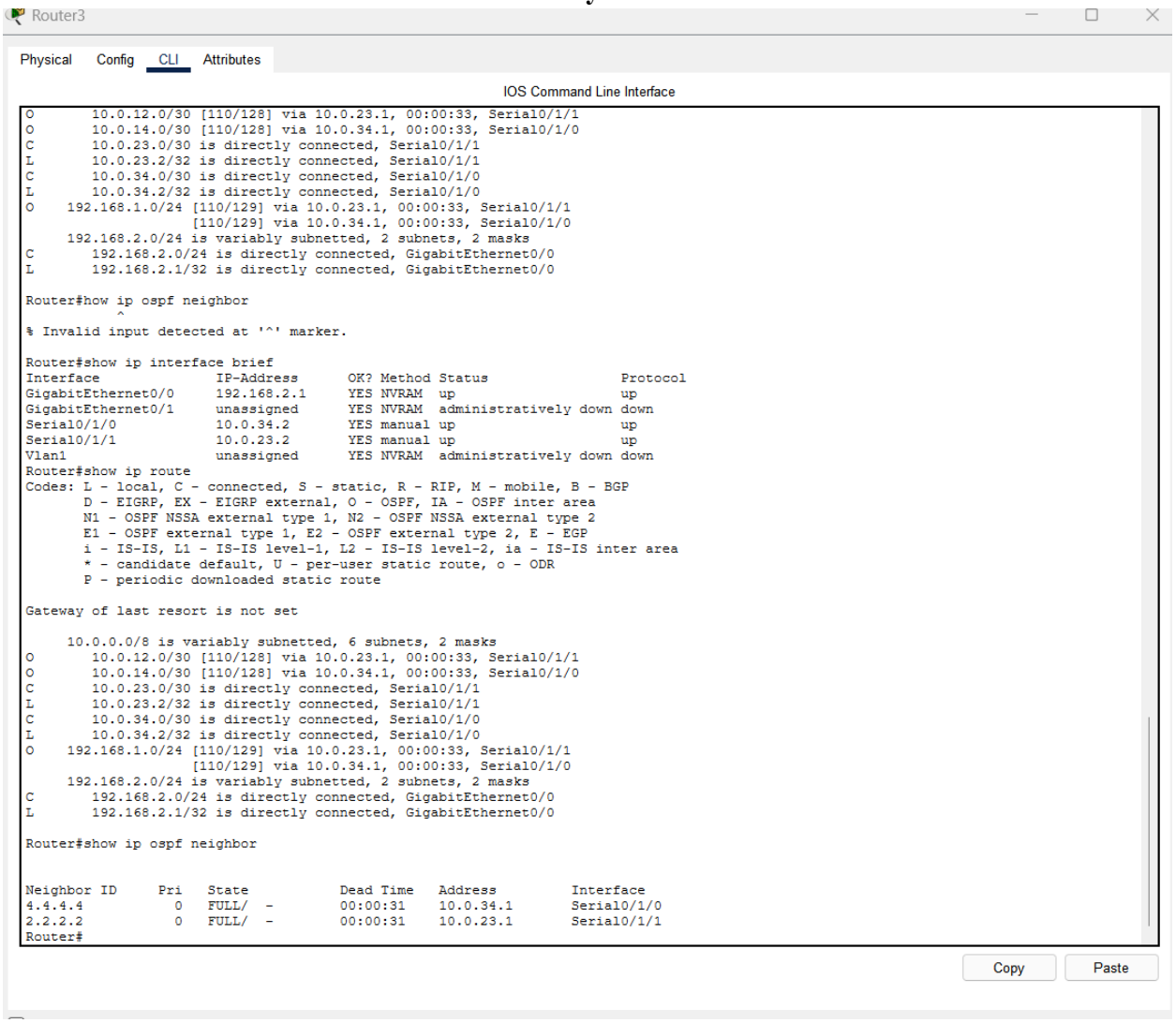
Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)

Electronics and Telecommunication Engineering Department

Academic year 2025-2026



Name: Chinmay Tawade

UID: 2024200132

Batch: B4

Branch : EXTC

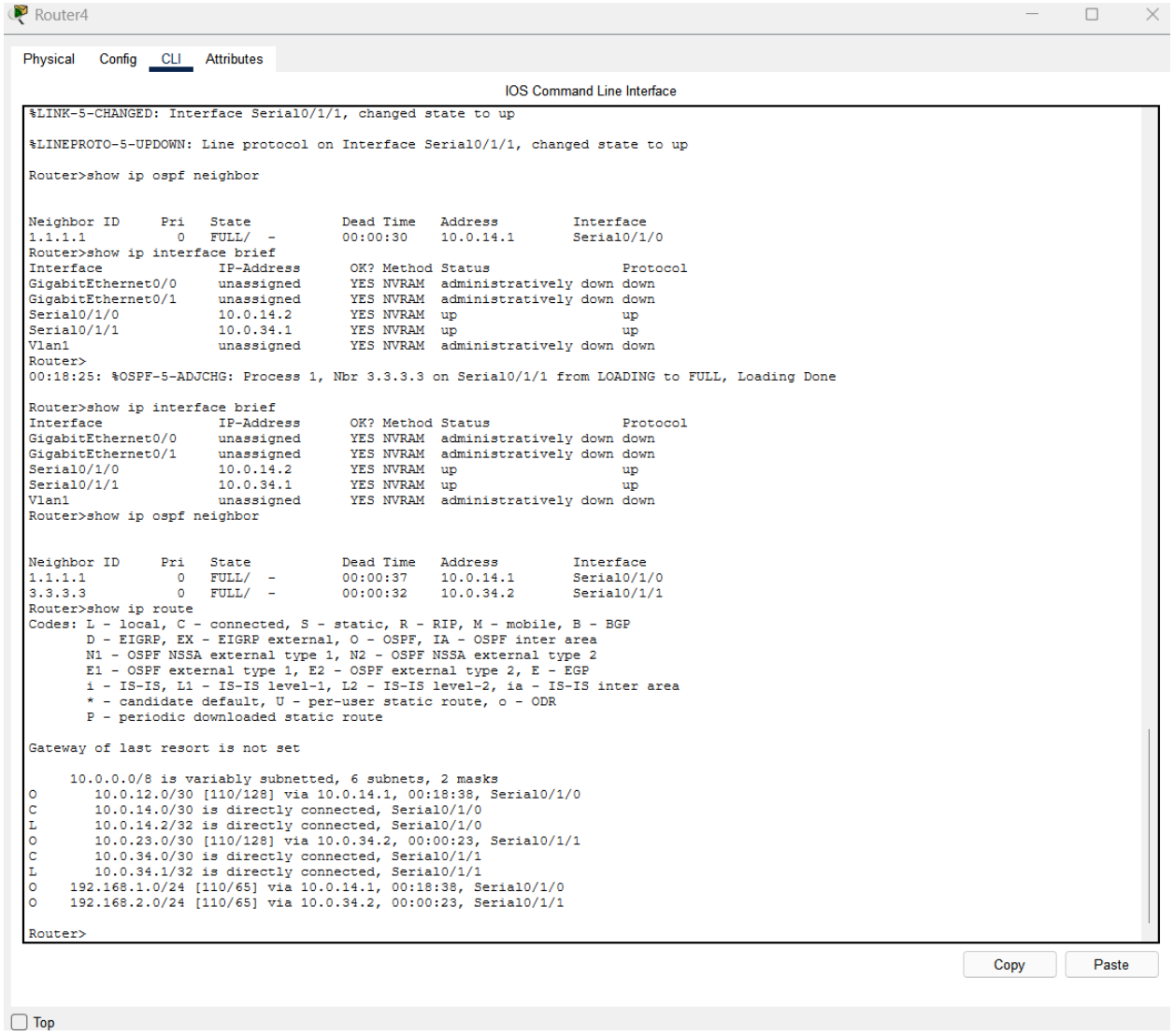
Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)

Electronics and Telecommunication Engineering Department

Academic year 2025-2026



Name: Chinmay Tawade
UID: 2024200132

Batch: B4
Branch : EXTC

Sardar Patel Institute of Technology

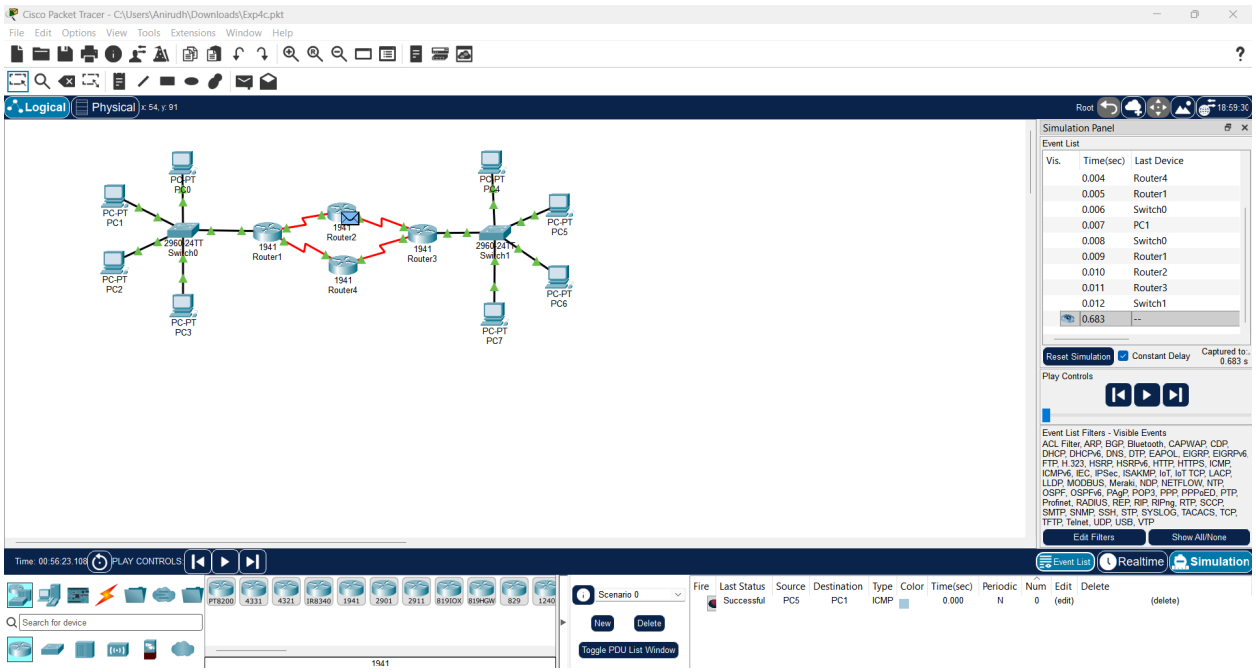
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)

Electronics and Telecommunication Engineering Department

Academic year 2025-2026

Sending Packets



Name: Chinmay Tawade

UID: 2024200132

Batch: B4

Branch : EXTC

Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Empowered Autonomous Institute Affiliated to University of Mumbai)

Electronics and Telecommunication Engineering Department

Academic year 2025-2026

Conclusion:

The experiment successfully showed how OSPF routing works in a network containing multiple routers. OSPF was configured on each router using the required network addresses and Area 0. Once the configuration was completed, the connected routers were able to discover each other and establish OSPF neighbor relationships.

The routers automatically exchanged routing information, and the routing tables were updated with routes learned through OSPF. This allowed devices on different networks to communicate without the need to manually configure routes on every router.

Overall, the experiment provided practical experience with dynamic routing and helped demonstrate how OSPF forms neighbor relationships and automatically shares and updates route information between routers.

Name: Chinmay Tawade
UID: 2024200132

Batch: B4
Branch : EXTC